



TECHNION |



The Henry and Marilyn Taub
Faculty of Computer Science

Planning and Reinforcement Learning in AI Systems

(PARLAI)

Lesson 2a - Actuation and Partial Observability

Sarah Keren

The Taub Faculty of Computer Science
Technion - Israel Institute of Technology

Accounting for Actions and Partial Observability

So far, we have considered systems that evolve over time.

These models describe **passive stochastic processes**.

Many real systems include an **agent** that can influence the system.

In this module, we extend the model in two ways:

- ➊ Adding **actuation**
- ➋ Accounting for **partial observability**

Actions, Rewards,
Policies

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Various Notations

This course fuses ideas from different research communities: learning, RL, planning, control, etc.

Each research community has its own set of notations. We will use various notations interchangeably.

Examples:

- state at time t can be represented as x_t or s_t
- observation at t is o_t or z_t .
- sometimes, evidence e_t is used to represent the sequence of observations, or observation-action pairs, or observation-action-rewards tuples.
- evidence $e_{1:t}$ collected over time is often called a *history* h .
- action at t is a_t or $\mathbf{u}_t$.

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Actions, Rewards, Policies

Passive Stochastic Dynamics

A stochastic dynamical system evolves according to some (possibly unknown) model

$$X_{t+1} \sim \text{Pr}(\cdot \mid X_t)$$

Characteristics

- The system evolves over time
- Transitions may be stochastic
- No agent controls the process

Examples:

- Weather evolution
- Population dynamics
- Electricity prices

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Adding Actuation

Now we have actions that can be applied at each time step

$$A_t \in \mathcal{A}$$

The system dynamics now also depend on the action:

$$\Pr(X_{t+1} \mid X_t, A_t)$$

The system is a **controlled stochastic process** or **controlled stochastic dynamical system**.

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Fixed Policy Assumption

This course is about coming up with *good* policies, but for now we assume actions are generated by a fixed policy π .

With a fixed policy, the challenge is to understand how the system evolves when the policy is followed.

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The system evolves according to

$$X_{t+1} \sim \Pr(\cdot \mid X_t, A_t)$$

with actions generated by the policy

$$A_t \sim \pi(\cdot)$$

The policy determines which actions occur, but the transition model determines their effect.

Actions can be defined in various ways:

- **Deterministic / non-deterministic / stochastic**
- **Continuous / discrete**
- From **high-level commands** (pick-up apple, make breakfast) to **low-level control signals** (joint torques, motor commands)
- May have **different durations or execution uncertainty**

- A **deterministic** policy is a mapping:
 - $\pi : \mathcal{S} \rightarrow \mathcal{A}$ from states to actions
 - $\pi : \mathcal{B} \rightarrow \mathcal{A}$ from beliefs to action
 - $\pi(s) / \pi(\beta)$ is the (single) action the agent will perform at state s or belief β .
- A **non-deterministic** policy is a mapping:
 - $\pi : \mathcal{S} \rightarrow \mathcal{A}^n$ from states to actions,
 - $\pi : \mathcal{B} \rightarrow \mathcal{A}^n$ from beliefs to actions
 - $\pi(s) / \pi(\beta)$ is the set of actions, one of which the agent will perform at state s or belief β .
- A **stochastic** policy is a mapping:
 - $\pi : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ from states to actions
 - $\pi : \mathcal{B} \times \mathcal{A} \rightarrow [0, 1]$ from beliefs to action
 - $\pi(s, a) / \pi(\beta, a)$ is the probability the agent will perform action a at state s / belief β .

Why do we need a **probabilistic** policy?

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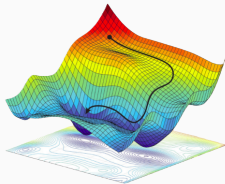
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Policy Parameters

- Instead of using a policy $\pi(a|s)$ that explicitly maps states (or beliefs to actions), we can use a **parameterized policy** $\pi_\theta(a|s)$, i.e., as a function with parameters θ .
- For deterministic policies θ specify the action taken at each state, and for stochastic policies, they induce a probability distribution over actions.
- Examples:
 - Linear policies $\pi_\theta(s) = \theta^\top s$ where θ is a vector of weights.
 - Neural network policies where θ are the weights of the network.



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Actions $\neq$ Agency

Introducing actions does not imply that an agent is making decisions.

- Actions can simply be **inputs applied to the system**.
- The actions may be generated by a **fixed controller or policy**.
- The challenge is to model how actions **influence the system dynamics** via $\Pr(X_{t+1} \mid X_t, A_t)$.

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A **reward function** assigns a numerical value to the outcome of an interaction with the environment.

Rewards are part of the **environment model** and describe the feedback received by the agent.

They play a central role in defining the **objective** that the agent will eventually try to optimize.

Representing Reward

There are various ways to represent reward.

With a **deterministic reward** $\mathcal{R}(s, a, s') \in \mathbb{R}$, once the transition $S_t = s, A_t = a, S_{t+1} = s'$ is fixed, the reward value is fixed.

In its most general form, the reward is a **stochastic variable**:

$$R_{t+1} \sim \mathcal{R}(\cdot \mid S_t = s, A_t = a, S_{t+1} = s')$$

Equivalently, the **reward distribution** can be written as

$$\Pr(R_{t+1} \mid S_t = s, A_t = a, S_{t+1} = s')$$

A deterministic representation of the **expected reward**:

$$R_{s,s'}^a = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a, S_{t+1} = s']$$

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Representing Reward (contd.)

Representing reward as a function

$$\mathcal{R}(s, a, s')$$

Often, rewards are defined over origin states $\mathcal{R}(s)$, destination states $\mathcal{R}(s')$, or over state-action pairs $\mathcal{R}(s, a)$.

Sometimes, rewards are induced by the environment (e.g., fuel costs), and sometimes they are designed (shaped) to encode how desirable an outcome is. This is called **Reward Shaping**.

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Reward Functions Examples

Robotics

$$\mathcal{R} = \begin{cases} +1 & \text{successful grasp} \\ 0 & \text{failure} \end{cases}$$



Games

$\mathcal{R}$ = score gained at step t



Control / Energy

$\mathcal{R}$ = -energy consumption



What is the reward function encoding for each setting?

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David Silver, Satinder Singh, Doina Precup, Richard S. Sutton
Reward is enough

Artificial Intelligence, Volume 299, 2021

<https://www.sciencedirect.com/science/article/pii/S0004370221000862>

<https://www.youtube.com/watch?v=u5hBHkbiHxA>

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From the Literature: Reward is enough?

Key claim: Many abilities associated with intelligence may arise from **maximizing reward signals**.

Examples of capabilities that have emerged from reward-driven learning:

- game playing (AlphaGo, AlphaZero)
- robotics control
- language models trained with feedback
- complex planning and strategy

A system that learns to maximize reward in rich environments may develop many components of intelligence.

Are there aspects of intelligence that cannot be captured by reward maximization?

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Return

Reward R : immediate feedback

Return G : total future reward

Key idea: future rewards are often **discounted** using a factor

$$0 \leq \gamma < 1$$

This reflects both:

- a natural preference for rewards received sooner rather than later, and
- a computational advantage, since discounting helps ensure convergence.

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- Expected return for episodic tasks:

$$G_t = R_{t+1} + R_{t+2} + R_{t+3} + \dots + R_T = \sum_{k=0}^T R_{t+k+1}$$

- Expected return for continuing (infinite) tasks:

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

- Returns at successive time steps are related:

$$G_t = R_{t+1} + \gamma(R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \dots) = R_{t+1} + \gamma G_{t+1}$$

Return quantifies the **total future reward**.

Value functions estimate how **good** it is:

- to be in a state
- to take an action in a state

Key idea: “goodness” is defined as **expected return**.

Since the rewards the agent can expect to receive in the future depend on what actions it will take, value functions are defined with respect to particular policies.

State-Value Function for Policy π :

$$V_{\pi}(s) \doteq \mathbb{E}_{\pi} [G_t \mid S_t = s]$$

State-Action-Value Function for Policy π :

$$Q_{\pi}(s, a) \doteq \mathbb{E}_{\pi} [G_t \mid S_t = s, A_t = a]$$

True vs. Estimated Values - Notations

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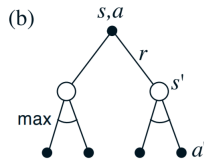
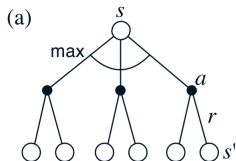
True (Unknown) value functions are denoted as:

- $V_{\pi}(s)$ — true state value
 - Defined via expectation: $V_{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid S_t = s]$
- $Q_{\pi}(s, a)$ — true action value
 - Defined via expectation: $Q_{\pi}(s, a) = \mathbb{E}_{\pi}[G_t \mid S_t = s, A_t = a]$

Estimated (learned) value functions (updated over time) are denoted as:

- $\hat{V}_{\pi}(s)$ — estimated state value
- $\hat{Q}_{\pi}(s, a)$ — estimated state-action value

Value Functions



Backup diagrams for $V_\pi(s)$ (left) and $Q_\pi(s, a)$ (right).

Since it may not be practical to keep separate averages for each state individually V_π and Q_π are often represented as parameterized functions (with fewer parameters than states).

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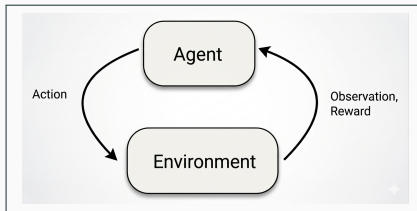
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The Agent-Environment Interface

The Agent-Environment Interface

An agent operates in the environment by taking actions.

An agent (and its behavior) is characterized by its **objectives** and **capabilities**.



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The Agent-Environment Interface

Typically, a model for AI decision-making comprises the following components:

- state space
- sensor function
- action space
- reward function
- horizon
- objective

A model is an **abstraction** of the underlying environment.

Each model can be used by different decision-making procedures.

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The Agent-Environment Interface

The agent interacts with the environment at discrete time steps $t = 0, 1, 2, \dots$

At each time step t , the agent observes the current state $S_t \in \mathcal{S}$ ¹ and selects an action $A_t \in \mathcal{A}(S_t = s)$.

As a result of applying A_t at S_t , at the next time step $t + 1$, the agent receives a numerical reward R_{t+1} and transitions to a new state S_{t+1} based on the dynamics model

$$S_{t+1} \sim \Pr(\cdot \mid S_t, A_t)$$

The agent and environment together thereby give rise to a sequence or **trajectory**: $\tau = S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3, \dots$

¹recall that we can use states as $\mathcal{X}$ instead

Actions, Rewards,
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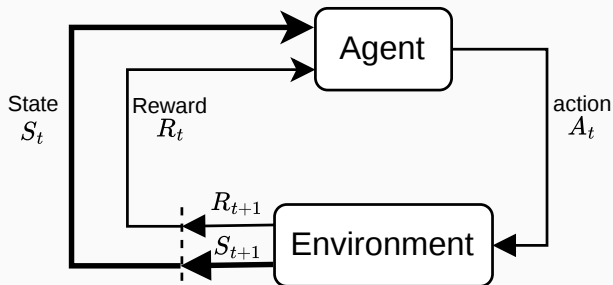
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The Agent-Environment Interface

- **Episodic tasks:** When the agent–environment interaction breaks naturally into sub-sequences (e.g., limited time or resources). Each episode ends in a special state called the *terminal state*.
- **Continuing tasks:** When the agent–environment interaction does not break naturally into identifiable episodes, but goes on continually without limit.

Real-world examples of episodic vs. continuing tasks ?

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Reactive Agents



SIMPLE ROOMBA

```
If BUMP = TRUE  
THEN Turn (random direction/amt)  
ELSE MOVE-STRAIGHT
```

<https://www.youtube.com/watch?v=7FSUtSurqA4>

We will focus on rational agents.

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Accounting for Partial Observability



What is the object?

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Accounting for Partial Observability



Now we can see an apple - but what is the state?

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From Full to Partial Observability

So far, we assumed the agent has access to the **true state**.

More precisely, **the agent observes all relevant features needed for decision-making**.

This assumption is called **full observability**.

In reality,

- the true state is often **hidden**
- the agent receives **partial and noisy observations**

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Instead of observing the state directly, the agent receives:

$$O_t$$

Observations are generated from the hidden state:

$$\Pr(O_t \mid X_t)$$

Examples:

- Robot camera observing object pose
- GPS measurement of position
- Medical test revealing hidden condition

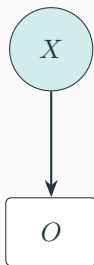
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Hidden state generates observations

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- The agent does *not* observe the true state X directly.
- Instead, it receives an observation O_t generated probabilistically from the hidden state via a **generative sensor model**:

$$O_t \sim \Pr(\cdot \mid X_t)$$

- It captures:
 - sensor noise,
 - limited sensing range or resolution,
 - ambiguity (multiple states producing similar observations).
- This model is called the **observation model** (or sensor model) $\mathcal{O}$.

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Partial Observability Over Time

$t = 3$



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Partial Observability Over Time

$t = 3$



$t = 15$



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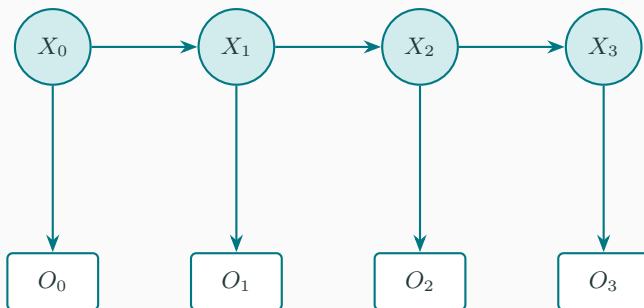
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Partial Observability Over Time

- When the environment is dynamic, over time, the agent observes only the **observation sequence**

$O_{0:T} = (O_0, O_1, \dots, O_T)$, while the **state sequence**

$X_{0:T} = (X_0, X_1, \dots, X_T)$ remains unobserved.

- At each time step, we now have **two** coupled processes:
 - **State dynamics:** $X_t \sim \Pr(\cdot \mid X_{t-1})$
 - **Observation generation:** $O_t \sim \mathcal{O}(\cdot \mid X_t)$

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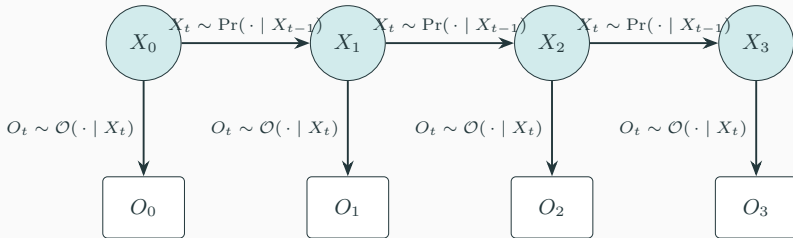
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Partial Observability Over Time

- Modeling assumptions:

- Markov dynamics:

$$\Pr(X_{t+1} \mid X_{0:t}) = \Pr(X_{t+1} \mid X_t)$$

- Memoryless observations:

$$\Pr(O_t \mid X_{0:t}, O_{0:t-1}) = \Pr(O_t \mid X_t)$$

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Partial Observability Over Time

- Modeling assumptions:

- Markov dynamics:

$$\Pr(X_{t+1} \mid X_{0:t}) = \Pr(X_{t+1} \mid X_t)$$

- Memoryless observations:

$$\Pr(O_t \mid X_{0:t}, O_{0:t-1}) = \Pr(O_t \mid X_t)$$

- Under these assumptions, the following conditional independencies hold:

$$X_{t+1} \perp\!\!\!\perp X_{0:t-1} \mid X_t, \quad O_t \perp\!\!\!\perp (O_{0:t-1}, X_{0:t-1}) \mid X_t$$

- Therefore, the joint distribution factors as:

$$\Pr(X_{0:T}, O_{0:T}) = \Pr(X_0) \prod_{t=0}^{T-1} \Pr(X_{t+1} \mid X_t) \prod_{t=0}^T \Pr(O_t \mid X_t)$$

This structure defines a **Hidden Markov Model (HMM)**.

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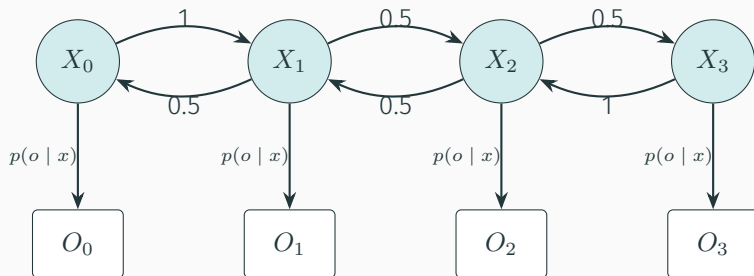
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Hidden Markov Model (HMM)



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Hidden Markov Model (HMM)

- An HMM is a **dynamic Bayesian network** with a repeated structure.
- Hidden states form a Markov chain:

$$\Pr(X_{t+1} \mid X_{0:t}) = \Pr(X_{t+1} \mid X_t)$$

- Each hidden state emits one observation:

$$\Pr(O_t \mid X_{0:t}, O_{0:t-1}) = \Pr(O_t \mid X_t)$$

- No direct dependence between observations.

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Acting Under Partial Observability

The Agent-Environment Interface Under Partial Observability

When acting in partially observable settings, the true state is **hidden**, and the agent relies on **observations** to estimate it.

As with the full observability, the agent interacts with the environment at discrete time steps $t = 0, 1, 2, \dots$

At each time step t , the system state is $S_t \in \mathcal{S}$ and the action performed is A_t .

The key difference is that the agent observes O_t instead of S_t via its sensor model $\mathcal{O}$.

The agent and environment together thereby give rise to a sequence or **trajectory**:

$$\tau = S_0, O_0, A_0, R_1, S_1, O_1, A_1, R_2, S_2, O_2, A_2, R_3, \dots$$

But the information available at time t is the **history**:

$$h = O_0, A_0, R_1, O_1, A_1, R_2, O_2, A_2, R_3, \dots$$

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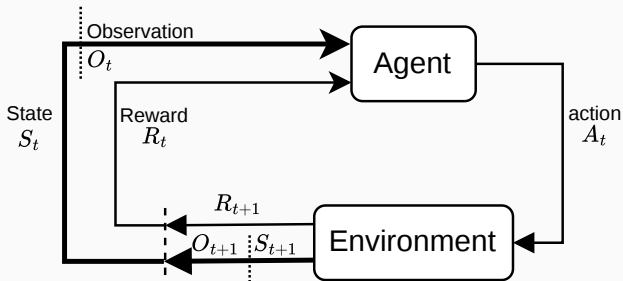
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The selection of A_t is based on the information collected thus far. Depending on the implementation, the decision at time t is over the history h_t or the belief β_t .

As a result of applying A_t at S_t , the agent receives at the next time step $t + 1$ a numerical reward R_{t+1} and transitions to a new state S_{t+1}

Importantly, regardless of whether the state is known or hidden, the dynamics are the same

$$S_{t+1} \sim \Pr(\cdot \mid S_t, A_t)$$

Maintaining a Belief

To track the hidden state, the agent maintains a **belief** (a probability distribution), also called the **belief state** when used for decision-making.

A belief:

- represents uncertainty over the true state
- summarizes the entire history
- serves as the agent's **internal state**

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Definition: Belief

Given a state space $\mathcal{X}$, a **belief** β_t at time t is defined as:

$$\beta_t(x) = \Pr(X_t = x \mid O_{0:t}, A_{0:t-1})$$

A belief is a probability distribution over states such that

$$\forall x \in \mathcal{X}: \quad \beta_t(x) \geq 0, \quad \sum_{x \in \mathcal{X}} \beta_t(x) = 1$$

Why is Belief Space Continuous and High-Dimensional

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Why is Belief Space Continuous and High-Dimensional

Belief Space: Beliefs are probability distributions over states:

$$\beta \in \Delta(\mathcal{X}) = \left\{ \beta \in \mathbb{R}^{|\mathcal{X}|} \mid \beta(x) \geq 0, \sum_{x \in \mathcal{X}} \beta(x) = 1 \right\}.$$

Continuous:

- Each state x gets a real-valued probability $\beta(x) \in [0, 1]$.

High-Dimensional:

- For $|\mathcal{X}| = n$ states, beliefs defined over $(n - 1)$ -dimensions.
- In many settings (e.g., robotics), the state space can be *continuous* $\Rightarrow$ belief is a *distribution over a continuous space*.

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Information States Beyond Beliefs*

Not all information states are probability distributions.

Examples:

- **Map representations:** occupancy grids, pose graphs, factor graphs
 - Occupancy grid:

I_t = probability map of free/occupied cells

- **Particle filter representation:**

$$I_t = \{X_t^{[i]}, w_t^{[i]}\}_{i=1}^N$$

- **Parametric approximations:**

- Kalman filter:

$$I_t = (\hat{x}_t, \Sigma_t)$$

- **Feature-based representation:**

$$I_t = f(o_{1:t})$$

- **Local information (multi-agent):**

$$I_t^i = (o_{1:t}^i, a_{1:t-1}^i)$$

- **Learned latent states:** RNN / GRU / LSTM embeddings
- **Full history (no compression):**

$$I_t = h_t$$

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In this course, we focus on beliefs.

Belief is a representational object.
How it is updated is a separate modeling choice.

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Definition: Belief Update

A **belief update** function $\mu : \mathcal{B} \times \mathcal{A} \times \Omega \rightarrow \mathcal{B}$ maps the current belief, action, and observation to the next belief:

$$\beta_t = \mu(\beta_{t-1}, a_{t-1}, o_t)$$

Filtering is the online inference task of maintaining the current belief by repeatedly applying belief updates as new observations arrive.

Bayesian Belief Update

Bayesian inference updates beliefs about a hidden state using observed evidence.

Bayes' Rule

$$\Pr(X | O) = \frac{\Pr(O | X) \Pr(X)}{\Pr(O)}$$

- **Prior:** $\Pr(X)$
- **Likelihood:** $\Pr(O | X)$
- **Posterior:** $\Pr(X | O)$
- **Evidence:** $\Pr(O)$ (normalization constant)

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What is missing for an agent acting over time?

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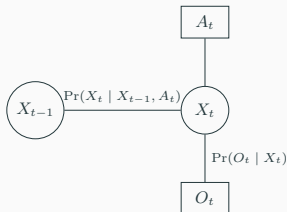
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Belief Update via Bayesian Filtering

Beliefs evolve over time as actions affect the state and new observations are received.

Bayesian filtering: combine dynamics (prediction) with observations (correction)



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Belief Update via Bayesian Filtering

Prediction (prior):

$$\bar{\beta}_t(X_t) = \sum_{X_{t-1}} \Pr(X_t \mid X_{t-1}, A_t) \beta_{t-1}(X_{t-1})$$

Correction (posterior):

$$\beta_t(X_t) \propto \Pr(O_t \mid X_t) \bar{\beta}_t(X_t)$$

$\propto$ is “proportional to”, i.e., normalized so probabilities sum to 1:

$$\beta_t(X_t) = \frac{\Pr(O_t \mid X_t) \bar{\beta}_t(X_t)}{\sum_{X'_t} \Pr(O_t \mid X'_t) \bar{\beta}_t(X'_t)}$$

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Algorithm 1 Bayes Filter

1: **Input:** β_{t-1}, A_t, O_t

2: **Output:** β_t

3: **for all** $X_t \in \mathcal{S}$ **do**

4: **Prediction (dynamics):**

$$\bar{\beta}_t(X_t) \leftarrow \sum_{X_{t-1} \in \mathcal{S}} \Pr(X_t \mid X_{t-1}, A_t) \beta_{t-1}(X_{t-1})$$

5: **Correction (observation):**

$$\beta_t(X_t) \leftarrow \Pr(O_t \mid X_t) \bar{\beta}_t(X_t)$$

6: **end for**

7: **Normalize:**

$$\beta_t(X_t) \leftarrow \eta \beta_t(X_t), \quad \eta = \frac{1}{\sum_{X_t \in \mathcal{X}} \beta_t(X_t)}$$

8: **return** β_t

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Bayes Filter (Continuous State Space)

Point probabilities $\Pr(\cdot)$ replaced by probability densities $p(\cdot)$.

Algorithm 2 Bayes Filter (continuous)

1: **Input:** β_{t-1}, A_t, O_t

2: **Output:** β_t

3: **for all** $X_t \in \mathcal{X}$ **do**

4: **Prediction (dynamics):**

$$\bar{\beta}_t(X_t) \leftarrow \int_{\mathcal{X}} p(X_t \mid X_{t-1}, A_t) \beta_{t-1}(X_{t-1}) dX_{t-1}$$

5: **Correction (observation):**

$$\beta_t(X_t) \leftarrow p(O_t \mid X_t) \bar{\beta}_t(X_t)$$

6: **end for**

7: **Normalize:**

$$\beta_t(X_t) \leftarrow \frac{p(O_t \mid X_t) \bar{\beta}_t(X_t)}{\int_{\mathcal{X}} p(O_t \mid X'_t) \bar{\beta}_t(X'_t) dX'_t}$$

8: **return** β_t

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Example: Bayesian Filtering for Localization

State: robot position $X_t \in \{Left, Right\}$

Initial belief:

$$\beta_{t-1}(Left) = 0.5, \quad \beta_{t-1}(Right) = 0.5$$

Transition model for noisy action *MoveR* (move right):

$$\Pr(Right|Left, MoveR) = 0.8, \quad \Pr(Left|Left, MoveR) = 0.2$$

$$\Pr(Right|Right, MoveR) = 0.9, \quad \Pr(Left|Right, MoveR) = 0.1$$

Observation model:

$$\Pr(O=Light|Right) = 0.9, \quad \Pr(O=Light|Left) = 0.3$$

(Complementary values are for no light)

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Example: Bayesian Filtering for Localization

Prediction:

$$\bar{\beta}_t(Left) = 0.2 \cdot 0.5 + 0.1 \cdot 0.5 = 0.15$$

$$\bar{\beta}_t(Right) = 0.8 \cdot 0.5 + 0.9 \cdot 0.5 = 0.85$$

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Example: Bayesian Filtering for Localization

Prediction:

$$\bar{\beta}_t(Left) = 0.2 \cdot 0.5 + 0.1 \cdot 0.5 = 0.15$$

$$\bar{\beta}_t(Right) = 0.8 \cdot 0.5 + 0.9 \cdot 0.5 = 0.85$$

Observation: sensor detects a Light

Correction (unnormalized):

$$\beta_t(Left) \propto 0.3 \cdot 0.15 = 0.045$$

$$\beta_t(Right) \propto 0.9 \cdot 0.85 = 0.765$$

Normalize:

$$\beta_t(Left) = 0.055, \quad \beta_t(Right) = 0.945$$

Key intuition: Action shifts belief $\rightarrow$ observation makes it

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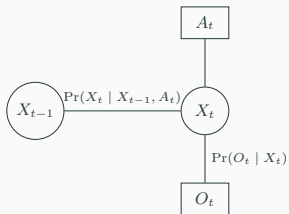
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A good representation of the state?

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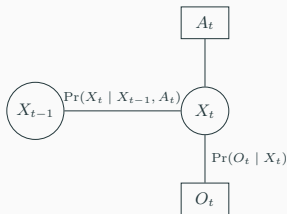
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A good representation of the state?

- Contains enough information for the agent to make 'good' decisions
- Can be updated efficiently
- Corresponds to 'natural' state variables in the physical world

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We extended stochastic dynamical systems in two ways:
Actuation and **Partial observability**

The agent maintains a belief $\beta_t(x)$ over the hidden state and makes decisions based on the belief.

Next, we will consider how agents **choose actions** to maximize performance.